

Introduction to Intellectual Property Law

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This one-day teaching module provides an introduction to U.S. intellectual property law, including trade secrets, utility patents, design patents, copyrights, and trademarks. It is designed for either the first day of a longer IP course or as a stand-alone module in a separate law, business, or policy class.¹

A. Why Intellectual Property?

“Intellectual property” or “IP” is an umbrella term for various legal rights attaching to creative intangibles. The kinds of intangible “information” or “knowledge goods” protected by IP rights are broad, ranging from books to pharmaceutical formulas to business plans to brand names like “Nike.” With the growth of the global knowledge economy, IP law has both risen in importance and adapted to ever-changing technologies, making it one of the most exciting and dynamic fields of legal practice today. And the stakes in IP cases are not just about which business will make the most money; for example, IP infringement can be a criminal offense; IP can incentivize—or limit access to—lifesaving technologies; IP is intertwined with cultural production and democratic discourse; and IP can reinforce—or subvert—existing power structures and inequalities.

Knowing the basics of modern IP law is valuable for every lawyer, business executive, and policymaker, and it *does not require a technical background*. It is also important to recognize that nothing about the current global IP system is inevitable, and to ask yourself at the outset: Why might it make sense for governments to create legal rights over creative intangibles? And why should these rights—which turn out to have very different legal details and goals—be lumped together under the “IP” umbrella? Let’s begin with the perspective of some private firms before turning to broader policy perspectives.

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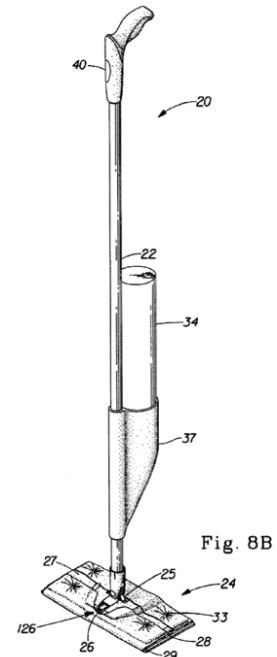
1. Business Perspectives

Imagine a firm with some new product—say, Procter & Gamble introducing the Swiffer WetJet mop. As illustrated at right, the Swiffer eliminates the need to clean the mop head or to use a separate bucket by using a disposable cleaning pad and an integrated system for delivering the cleaning solution, which can be sprayed by pressing a button on the handle. The Swiffer was developed by a large R&D team at significant time and expense, including many hours recording and analyzing prior floor-cleaning methods. (This is actually true!)

Now imagine what would happen if P&G started selling the Swiffer in a world without any form of IP. If the Swiffer was successful, competitors could swoop in, copy the product, and sell it at a lower price. They would not have to sell it at a high enough price to recover their initial R&D expenditures. (Think about how much cheaper it usually is to buy the generic version of a drug than the brand-name version.) P&G would have to lower its price just to stay competitive. And at this lower price, it might never earn enough profit to make up for its own initial R&D costs. With the looming threat of being undersold like this, what firm would want to invest in R&D in the first place, knowing that it might never recoup its investment?

IP law can help P&G avoid this problem by prohibiting P&G's competitors from copying many aspects of the Swiffer—including possibly the Swiffer itself. P&G could use trade secret law to protect proprietary business information, including not just technical information like how to manufacture the mop, but also lists of suppliers and customers, operating procedures, and marketing plans. P&G could obtain patents on any new aspects of the product—for instance, if the Swiffer were the first mop to integrate a cleaning spray, that might be patentable. P&G could protect the design of the Swiffer product with design patents, copyright, and as “trade dress” under trademark law. It could protect the content of the Swiffer instruction manual with copyright law. And of course, P&G would want to claim trademark rights in the names “Swiffer” and “WetJet.” All of these measures would bestow business advantages on the Swiffer. P&G might then be able to sell it for a higher price than competing products, or it might even allow P&G to block competing products from entering the marketplace at all (if P&G's patents are strong). IP protections might thus enable P&G to recoup its initial R&D investment and more.

A savvy and well-advised company will consider all of these issues from the outset. P&G's optimal IP strategy will depend on its business goals, but it should consider questions such as the following:



- Have reasonable efforts been taken to protect trade secrets, such as having employees and supply chain partners sign non-disclosure agreements (NDAs)?
- Has P&G ensured that all of its employees have signed IP assignment agreements and that no other parties—such as former employers of the Swiffer inventors—have a viable ownership claim to relevant IP?
- Should P&G file provisional patent applications as early as possible? How much is it worth spending on patent protection? Is it worth filing for patents outside the United States?
- Is it worth conducting a freedom to operate search for other patents that P&G may need to design around or license?
- Is the marketing team coordinating with legal counsel to set up a strong trademark strategy, including checking for products that already use similar names or designs, and considering trademark protection when choosing its own names and designs?
- If other firms appear to be infringing any of P&G's IP rights, what is the best business response? (Just because you have a plausible legal claim doesn't mean it's always a good idea to send a demand letter or file a lawsuit!)

Now consider another firm that wants to make a related product—say, EcoMop, a (fictitious) startup that wants to sell an eco-friendly Swiffer alternative that uses reusable and machine washable cleaning pads rather than the disposable pads used by the Swiffer, along with a proprietary biodegradable cleaning solution. Competing with the Swiffer may involve some legal risk, especially if P&G enforces its IP aggressively, but good lawyering does not fixate blindly on risk. In helping EcoMop evaluate the benefits and costs of different business strategies, EcoMop's lawyers will have their own IP questions to consider:

- Can EcoMop hire a former P&G employee who has experience with Swiffer manufacturing or marketing without undue legal risk?
- Can EcoMop manufacture a mop that doesn't infringe any of P&G's patents? If not, can it invalidate or license those patents? Should it consider a partnership with P&G?
- How different does the design need to be to avoid infringing design-related IP? Can EcoMop copy aspects of the design that are important to the mop's functionality and consumer appeal?
- Can EcoMop market its product as the "EcoMop Wet Jet"? As "An Eco-Friendly Alternative to Swiffer"? Can it paraphrase the Swiffer instruction manual?
- How should EcoMop protect its own intellectual property?

Can you see why each of these questions is important to each firm's strategy? Can you think of other IP questions that these firms might want to consider? Keep these questions in mind as you read through the overview of the different types of IP law below. But first, it is worth considering why *governments* would grant any of these IP rights in the first place. How P&G and EcoMop (and their other competitors) use IP law has tremendous *private* importance to these firms, but why should *society* care which mop manufacturer makes the most money?

2. Policy Perspectives

From a social welfare perspective, it doesn't matter which mop company makes more money—but it does matter whether innovative new mops are available, especially if those mops improve the lives of consumers. As noted above, without IP law, copycat firms could immediately enter the Swiffer market at a lower price. This competition would be good for consumers who could buy Swiffers more cheaply. But if P&G anticipates that its market is likely to be undercut such that it will not be able to recover its R&D costs, then it wouldn't invest in R&D in the first place! Innovative new products—like the Swiffer—might never be created and never made available to the public. Incentivizing individuals and firms to invest in creating, disclosing, and distributing new knowledge goods is one of the core justifications of U.S. IP law.

In economic terms, the main problem IP addresses is that knowledge is often a “public good”—many people can benefit from the idea of an integrated mop without decreasing its benefit for others (it is “nonrivalrous”), but once P&G discloses this idea, it is difficult to prevent others from using the idea (it is “nonexcludable”). If P&G can't exclude others from using the information, then it won't be able to charge them money from using it, which means that it won't benefit from others' uses of the information. And the fact that other people or firms might benefit from the information P&G developed won't make P&G more likely to develop that information in the first place—P&G cares only about its own bottom line. In economic terms, P&G won't account for benefits that it cannot internalize when deciding how much to invest in mop R&D—which might mean that the Swiffer never gets developed at all. Public goods will generally be underproduced by the market. This creates a justification for government intervention in the form of IP laws. IP laws convert information from a public good into a private good by allowing its owner to exclude others from using it.

But this *doesn't mean that more IP is always better* for society! IP rights create real social costs, including the cost to consumers who pay much more for Swiffers than their marginal cost of production, the cost to P&G of finding and licensing the IP of earlier innovators, the cost for subsequent innovators like EcoMop who want to improve on the Swiffer design, and the administrative and transaction costs of obtaining, licensing, and litigating IP rights. Indeed, many aspects of the Swiffer actually had been invented before, and P&G had to license technology from the Japanese firm Unicharm before marketing the Swiffer. To determine the optimal IP

rights for mops, a policymaker might want to know: How socially valuable are new mops like the Swiffer? What IP rights are actually needed for their development? Are there less costly policy tools that could achieve the same goal? Innovation will still occur without formal IP, so policymakers should consider whether IP causes enough *additional* innovation to be worth the costs.

Of course, IP policymakers aren't just focused on mops. How might optimal IP policy differ if we focus instead on vaccines? Movies? Software? Fashion? Should it matter how expensive or time-consuming it is to bring a new product to market, or how cheap it is to copy? Does it make sense to have patents on everything from semiconductors to business methods, and copyrights on everything from fine art to software, rather than creating specific types of IP rights that cover specific types of technology? What are the benefits of *not* trying to tailor IP law by technology area?

For that matter, what are the benefits of using IP compared with non-IP policies to incentivize the creation, disclosure, and distribution of knowledge goods? Policymakers can fund creative production directly, such as through the National Institutes of Health or the National Endowment for the Arts. They can offer prizes for successful innovations, such as the prizes offered at Challenge.gov or the market subsidies provided by government health insurance programs like Medicare. They can use tax incentives, such as the R&D incentives of I.R.C. §§ 41, 174 or the tax benefits most states offer for film production. Compared with government-tailored incentives, IP has the virtue of rewarding creators based on how much consumers actually value their creations, rather than leaving this choice to government bureaucrats—but can you think of instances when this virtue is really a downside? In practice, U.S. innovation policy mixes and matches these policy tools, including by sometimes using IP as an incentive while allocating access to the resulting innovations at low or no cost to patients, as under the Medicaid system. On the benefits of this pluralistic approach, see Daniel J. Hemel & Lisa Larrimore Ouellette, *Innovation Policy Pluralism*, 128 Yale L.J. 544 (2019).

One important dimension for comparing IP and other innovation policies is through the lens of innovation and inequality, which is an area of growing policy interest. Wide disparities exist—including by race, gender, and socioeconomic status—in who creates IP, who has access to IP-protected products, whom those products are made for, and who financially benefits from IP. These disparities raise concerns for both equity and economic growth. For example, economists estimate that if women, racial minorities, and people from low-income backgrounds invented at the same rate as white men from families in the top income quintile, the rate of innovation in America would quadruple. See Alex Bell et al., 134 Q.J. Econ. 647 (2019). How do you think the choice to use IP law rather than alternative policy instruments affects these disparities? When policymakers consider the social value created by innovation, should they give greater weight to the interests of the worse off (however defined)? How should innovation policy consider tradeoffs between total welfare improvements

and equity objectives? What about issues of *global* equity—how should U.S. policymakers weigh the substantial benefits of innovation beyond our borders?

Finally, although most justifications for U.S. IP law focus on overall social welfare rather than the benefits for individual innovators or creators, a long line of literature still emphasizes natural rights theories of IP, particularly in continental Europe. For example, John Locke's 17th century theory of property—that people are entitled to the fruits of their labor, “at least where there is enough, and as good, left in common for others”—has been applied to intangible property. What does a Lockean approach imply about situations in which two people independently labored to make the same creation? Does this theory make more sense for some kinds of IP than others? For a Lockean argument for a minimalist copyright regime, see Mala Chatterjee, *Lockean Copyright Versus Lockean Property*, 12 J. Legal Analysis 136 (2020). Other scholars have argued that intellectual creations should be protected because they are “closely bound up with [creators'] personhood.” Margaret Jane Radin, *Property and Personhood*, 34 Stan. L. Rev. 957 (1982). Do you think any of these natural rights theories are more compelling for some types of creations than others?

These competing theories of IP are not just of academic interest. Policymakers regularly adapt IP laws based on policy concerns. Congress considers many amendments to IP statutes. Courts interpret those statutes, often through a policy-driven common-law approach. Administrative agencies like the U.S. Patent and Trademark Office issue new regulations and make decisions on whether a given IP right should be granted. And even if you represent a private firm, considering how the firm's interests align with broader policy goals is worthwhile: many important recent changes to IP law have been driven by private firms with lawyers who can make an effective case for why a given change serves public interests more broadly.

B. Types of Intellectual Property

This section provides an overview of five areas of U.S. IP law—trade secrets, utility patents, design patents, copyrights, and trademarks—as well as a summary of remedies across IP and the governing international framework. Of course, this brief introduction does not cover all the nuances of these doctrines, and there are additional legal protections for intangible property beyond those covered here, including rights of publicity, express or implied contractual protections for data and ideas, and rights attaching to specialized regulatory regimes such as approval of new drugs by the Food and Drug Administration (FDA). But based on this overview, what questions do you think P&G and EcoMop should consider related to each area of IP? What policy goals does each area appear to serve?

1. Trade Secrets

Trade secret law protects confidential business information, ranging from the Coca-Cola recipe to technical details about a new manufacturing process to marketing plans and customer lists. Note that firms can use secrecy even without trade secret law; for example, the Coca-Cola recipe can be kept in a secure vault with limited access. But by providing companies with a legal cause of action against parties who “misappropriate” their secrets, trade secret law can encourage innovation, limit the secrecy arms race, and encourage efficient disclosure and licensing. Trade secret law is also influenced by tort-based deterrence theories grounded in industry norms and good faith.

Trade secrets also have tremendous economic and social importance. Survey evidence indicates that firms generally value trade secrets more highly than patents for protecting their investments in innovation. And trade secret litigation can have enormous stakes, including financial liability in the millions or billions of dollars and potential criminal sanctions. Trade secret law also implicates important questions of social policy beyond promoting innovation. The vast majority of cases involve former employees and thus raise questions about the rights of employees to change jobs or to disclose information about the workplace. Trade secret law is also used to protect information of public interest, ranging from firm diversity practices to the algorithms used for convicting and sentencing criminal defendants to the information considered by the FDA when deciding whether to approve a new drug or vaccine.²

U.S. trade secret law evolved as a matter of state common law, but every state except New York has now enacted a trade secret statute based on the Uniform Trade Secrets Act (UTSA), with the specific language and interpretation varying by state.³ The 2016 Defend Trade Secrets Act (DTSA) also provides a federal civil cause of action modeled on the UTSA. The DTSA doesn’t preempt state law, so trade secret litigation typically involves claims under both the UTSA and DTSA. In addition, trade secret misappropriation may be criminally punished under the federal Economic Espionage Act of 1996.

The definition of a “trade secret” is provided by UTSA § 1(4) and by the DTSA at 18 U.S.C. § 1839(3). Under both, establishing a trade secret involves three elements:

² For additional reading on these issues, see Jamillah Bowman Williams, *Diversity as a Trade Secret*, 107 Geo. L.J. 1685 (2019); Rebecca Wexler, *Life, Liberty, and Trade Secrets: Intellectual Property in the Criminal Justice System*, 70 Stan. L. Rev. 1343 (2018); and Allison Durkin et al., *Addressing the Risks That Trade Secret Protections Pose for Health and Rights*, 23 Health Hum. Rights 129 (2021).

³ The UTSA is at <https://www.wipo.int/edocs/lexdocs/laws/en/us/us034en.pdf>. It has no legal force on its own, so you should always check (and cite to) the relevant state statute. North Carolina is the only state whose statute is not explicitly based on the UTSA, but it is very similar. New York’s trade secret law remains heavily influenced by the 1939 *Restatement of Torts*.

1. The information must be *secret*, meaning “not . . . generally known to, and not . . . readily ascertainable” by other persons. (Some states, such as California, do not have the “not readily ascertainable” language.) If the information is not secret, even egregious theft cannot support a trade secret claim (though it may be independently illegal).
2. The information must have “independent economic value” due to its secrecy. This is usually easily established; why would the information have been misappropriated if it wasn’t valuable?
3. The owner must have taken “reasonable” efforts to maintain its secrecy. This may be assessed through an economic cost-benefit analysis, comparing the cost of precautions to the value of the secret. Reasonable efforts may include precautions such as non-disclosure agreements and physical or electronic access limitations.

This definition is very broad! Trade secrets can include business plans, internal operating procedures, customer and supplier lists, financial information, technical processes, schematics, computer programs, unpublished documents, and any other information of value.

Trade secrecy “misappropriation” is defined by UTSA § 1(1)–(2) and by the DTSA at 18 U.S.C. § 1839(5)–(6). It includes acquisition, disclosure, or use of a trade secret by “improper means,” which “includes theft, bribery, misrepresentation, breach or inducement of a breach of a duty to maintain secrecy, or espionage through electronic or other means.” Misappropriation generally involves independently unlawful conduct, such as breach of contract, trespass, or violation of the Computer Fraud and Abuse Act, but a comment to the UTSA also specifies: “Improper means could include otherwise lawful conduct which is improper under the circumstances; e.g., an airplane overflight used as aerial reconnaissance to determine the competitor’s plant layout during construction of the plant. *E. I. du Pont de Nemours & Co., Inc. v. Christopher*, 431 F.2d 1012 (5th Cir. 1970).” The majority of trade secret misappropriation cases do not involve this kind of corporate espionage; rather, they involve violation of an express or implied confidential relationship, such as disclosure or use of trade secrets by a former employee.

Trade secrecy protection can last forever, but it is lost once the information is disclosed, including by third parties who have independently invented or reverse-engineered the secret. Thus, the choice of whether to patent an invention or protect it as a trade secret may be informed by comparing the patent term with the expected lifespan of the secret. In practice, inventors can often patent one aspect of an invention while maintaining trade secrecy protection over any aspects that are not disclosed in the patent application. Also, patent applications can be protected as trade secrets before they are published.

For lawyers, trade secret law can be a rewarding practice area because it is geographically flexible, it often involves substantial client counseling and corporate work in addition to litigation, and there are many major open questions that still need to be resolved by the courts. There is also overlap with patent practice because many lawsuits involve both patent infringement and trade secret misappropriation claims.

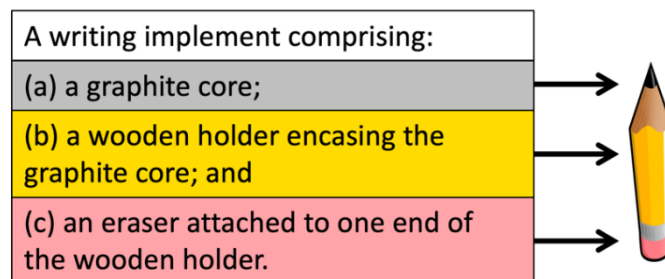
2. Utility Patents

Utility patents are rights granted by the U.S. Patent and Trademark Office (USPTO) for new and useful inventions, which can range from new pharmaceuticals to software algorithms. U.S. patent law is grounded in the Constitution's IP Clause, at Article I, Section 8, Clause 8, which allows Congress "[t]o promote the progress of science and useful arts, by securing for limited times to authors and inventors the exclusive right to their respective writings and discoveries." In other words, inventors are granted temporary rights to encourage them to create new inventions.

Importantly, a patent doesn't provide an affirmative right to make the invention it describes; rather, the owner of a patent can *prevent others* from making, using, selling, offering to sell, or importing the invention during the life of the patent, which typically lasts 20 years from when the patent application is filed. In practice, patents rarely confer a true monopoly, and making a patented product may require numerous licenses from owners of earlier patents.

Patent law—both the utility patents discussed here and the design patents described below—is exclusively federal. The Patent Act is codified at Title 35 of the U.S. Code, and federal courts have exclusive jurisdiction over patent cases. (In general, state courts are courts of general jurisdiction and can hear federal claims, but not for patents.) In addition, all appeals of patent cases go to the Court of Appeals for the Federal Circuit in D.C., which is unique among the thirteen federal circuit courts in having a jurisdiction based entirely on subject matter rather than geography.

Also unlike other areas of IP, for which some IP rights exist without any special formalities, patent rights depend on having a patent granted by the USPTO. Every patent has *claims* that define the boundaries of the legal right. For example, a simple patent claim for a pencil might read as follows:



This claim is only valid if it satisfies stringent legal requirements. Under 35 U.S.C. § 102 and § 103, it must be *novel* and *nonobvious* when compared with everything available to the public that preceded it, including earlier publications, patents, and other inventions that have been sold or used. “Novel” means that the invention is genuinely new; it has never before existed. “Nonobvious” means that someone skilled in the relevant technological field wouldn’t have been able to easily create the invention by combining two or more pre-existing things. The collection of information available to the public before a patent application is filed is known as the *prior art*. For example, our pencil claim might be rejected as an obvious combination of prior art erasers and wooden cylinders with graphite cores. These requirements are evaluated from the perspective of the Person Having Ordinary Skill in the Art (PHOSITA), who you can think of as a hypothetical “reasonable researcher” in the field of the patent at issue.

Under 35 U.S.C. § 112, the patent document also must show that the claim satisfies the requirements of *enablement* (teaching the PHOSITA how to make and use the invention without “undue experimentation”) and *written description* (showing that the inventor “possessed” the invention at the time of filing). These requirements help prevent inventors from receiving legal rights broader than what they actually contributed. The inventor of a wooden pencil cannot receive a patent claim covering the future development of pens unless she is able to teach other skilled artisans how to make and use pens as well.

In addition, a patent claim must satisfy the *patentable subject matter* requirement of § 101, which prevents inventors from claiming a broad “abstract idea” or “law of nature.” For example, an inventor cannot claim the general idea of taking payments electronically over the internet without doing something more, such as improving the technical functioning of the computer system. Patentable subject matter doctrine was revitalized by the Supreme Court in cases including *Alice v. CLS Bank*, 573 U.S. 208 (2014), and *Mayo v. Prometheus*, 566 U.S. 66 (2012), and it is now used to invalidate many claims in certain technical areas—such as software and financial patents—early in litigation, although its contours remain unclear and contentious.

The owner of a valid patent claim can prevent others from directly exploiting the invention—including by making, using, or selling it—or from indirectly inducing or contributing to infringement by others. Determining whether a claim is infringed involves comparing the allegedly infringing product or process with the text of the claim. This analysis typically requires *claim construction*, which involves determining what a PHOSITA would think the technical language of the patent claim actually covers. For example, does a pencil made from a wood-plastic composite infringe a claim covering pencils with a “wooden holder”?

Defendants in patent cases tend to focus on arguing that a claim is invalid or not infringed because they have few other options. It typically doesn’t matter if the

infringer was unaware of the patent or was using the invention only for research—ignorance is no defense. There is also no patent equivalent to copyright fair use (discussed below).

3. Design Patents

When most people say “patent,” they mean a utility patent, or a patent on a useful invention. But there are two other kinds of patents granted in the United States: *design patents* on new ornamental designs and *plant patents* on certain new plant varieties. In 2020, the USPTO granted 352,049 utility patents, 34,877 design patents, and 1,398 plant patents. This overview doesn’t cover plant patents, although they can be quite important in the agricultural industry.⁴ The lifespan of a design patent is 15 years from the date of grant (or 14 years for design patents filed before May 13, 2015).

FIG.1



Like utility patents, design patents only protect inventions that are novel and nonobvious relative to the prior art. Small modifications are often enough to clear these bars,⁵ although the Federal Circuit recently made it easier to invalidate design patents as obvious in *LKQ v. GM*, 102 F.4th 1280 (Fed. Cir. 2024) (en banc). Disclosure and claiming requirements for design patents are satisfied through drawings, such as the picture at right of the ubiquitous Aeron chair. Design patents must be “ornamental,” but this is also not an onerous requirement: a design patent will be invalidated for ornamentality only if the design is “dictated by” its function. The requirements for design patents are contained in 35 U.S.C. § 171.

The relative ease of acquiring design patents comes with a correspondingly narrow scope: design patents are infringed only if an ordinary observer familiar with the prior art would consider the two designs to be “substantially the same.” That is, the accused product (or relevant portion thereof) must look the same as the claimed design as a whole. For example, Apple received an award of over \$500 million because the jury considered Samsung smartphones and tablets to have substantially the same design as Apple claimed in its design patents.

⁴ Plants can also be protected with utility patents, trade secrets, and with another form of IP, *plant variety protection certificates*. Plant names can also be protected as trademarks. For more information on plant patents and design patents, see Sarah Burstein, Sarah R. Wasserman Rajec & Andres Sawicki, *Patent Law: An Open-Access Casebook* (2022).

⁵ For an illustration, scroll through some of these 10,000+ recent U.S. design patents on chairs: <https://patents.google.com/?q=chair&country=US&after=priority:20000101&status=GRANT&type=DESIGN>

4. Copyrights

Copyright law protects a wide variety of creative expression against copying, including not only works that might be the subjects of literature, art, and film classes, but also—more controversially—product designs and computer programs. Like patent law, copyright law is grounded in the Constitution’s IP Clause. But the “limited times” for copyrights are far longer than for patents: 70 years after the death of the author or, for corporate works, 120 years after creation or 95 years after publication, whichever is shorter.

The primary source of U.S. copyright law is the federal Copyright Act at Title 17 of the U.S. Code, although there are also limited state law protections for creative works not protected by federal law. You don’t need a formal registration or a © notice to have a valid copyright, although registering a work with the U.S. Copyright Office at Copyright.gov provides important benefits. For example, copyright registration is required before filing most infringement suits, can serve as *prima facie* evidence of validity, and is a prerequisite for statutory damages and attorneys’ fees.

Under 17 U.S.C. § 102, copyright protects “original works of authorship,” which courts have interpreted to require both a *modicum of creativity*—a much lower bar than patent law’s nonobviousness requirement—and *independent creation*. If you take a photograph that happens to look strikingly similar to an earlier picture that you weren’t aware of, you can copyright your photograph (because of the minimal creativity involved in arranging the photo composition) and your photo doesn’t infringe the earlier photographer’s copyright. A copyrighted work must also be “fixed in any tangible medium of expression,” meaning that some embodiment “is sufficiently permanent or stable to permit it to be perceived, reproduced, or otherwise communicated for a period of more than transitory duration.” §§ 101–102. This broad language is intended to cover future technological development: works can be fixed using canvas, paper, sculpture, records, DVDs, MP3s, and a variety of other digital or nondigital formats.

Copyright law doesn’t protect ideas—only expressions of ideas. Copyright thus cannot cover useful methods (that’s for utility patent law), whether a method of accounting, a cooking recipe, or a functional computer algorithm. But it can protect a particular creative expression of any of these ideas, such as the text of a book describing how to use the accounting method, a cookbook author’s asides about the memories a recipe evokes, or software in which a programmer has made one of many choices in how to express an algorithm in code. Of course, the dividing line between idea and expression can be murky. *See Google v. Oracle*, 141 S. Ct. 1183 (2021) (sidestepping the issue of whether the computer code at issue was copyrightable).

Copyright also doesn’t protect facts, such as a factual database. (Other countries offer *sui generis* IP protection for databases.) If after painstaking archival research you develop a new theory about how a legal statute should be interpreted

and you write a seminar paper describing your ideas about the facts you have uncovered, copyright law would not prevent another student from copying your ideas and facts without attribution if they do not copy the creative expression you used. (However, they will have committed plagiarism and may be sanctioned for academic misconduct!) Although facts are not themselves copyrightable, you can receive copyright protection for a *compilation* of facts (or other works, as in a cookbook or poetry anthology) if there is minimal creativity in your choice of the *selection* and *arrangement* of the facts.

You also generally cannot copyright a “useful article”—one with “an intrinsic utilitarian function”—unless its design has aesthetic features that can be conceptually separated from its utilitarian aspects. 17 U.S.C. § 101. For example, you cannot use copyright law to protect the design of a simple utilitarian shovel or T-shirt—this is the domain of utility patent law. But you can copyright a shovel with elaborate handle carvings or a T-shirt with a creative graphic design. As with the idea/expression dichotomy, the dividing line has been contentious, and recent Supreme Court intervention in a case that upheld the copyrightability of cheerleader uniforms has done little to resolve the issue. *Star Athletica v. Varsity Brands*, 137 S. Ct. 1002 (2017).

A valid copyright provides protection against reproduction, adaptation into a “derivative work” (like a translation, movie version, or sequel), distribution, performance, or display. This protection extends not only to exact copies, but also to substantially similar variations, such as copying part of a song’s melody for another song. Establishing an infringement claim requires showing (1) that the infringing work was actually copied from the original rather than independently created, and (2) that the works have “substantial similarity” from the perspective of an ordinary consumer. Before the works are compared for similarity, it is important to “filter out” any unprotected elements, such as abstract ideas, facts, and public domain materials.

Copyright protection has many statutory limitations and compulsory licensing regimes, but the most prominent limitation is the doctrine of fair use under 17 U.S.C. § 107. If you are engaged in what the law considers “fair use” of a copyrighted work, you are not infringing and cannot be made to pay the copyright owner. Use of a copyrighted work is more likely to be fair use when the use is noncommercial or for critique or research, when the copyrighted work is factual, when only a small portion of the work is used, and when the use doesn’t supplant potential markets for the copyrighted work. Under the Supreme Court’s interpretation of the fair use statute, the most important factor has become whether the “purpose and character” of the use is “transformative”: whether it “adds something new, with a further purpose or different character, altering the first with new expression, meaning, or message.” The Supreme Court relied on this transformativeness inquiry in holding Google’s repurposing of certain computer software owned by Oracle to be fair use. *Google v. Oracle*, 141 S. Ct. 1183 (2021). But fair use cases are rarely clear-cut. Most recently, the Supreme Court held that the “purpose and character” factor did not favor fair use in the context of a portrait of the musician Prince by pop artist Andy Warhol, but the

impact of this case will not be clear for years to come. *Andy Warhol Foundation v. Goldsmith*, 143 S. Ct. 1258 (2023).

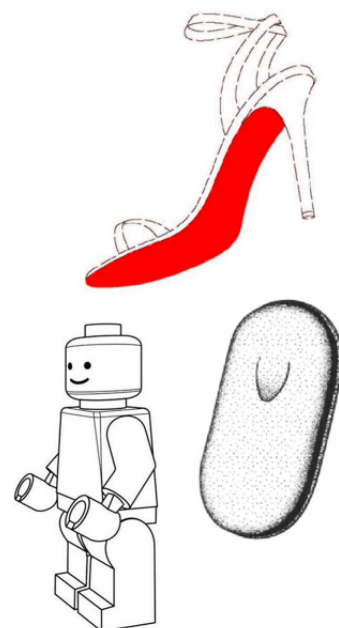
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5. Trademarks

A trademark is something that identifies the source of a good or service. Unlike patent, trade secret, and copyright law, trademark law is not generally intended to provide incentives for innovation. Rather, trademark law is meant to help consumers by improving the quality of information about the source of goods and services. This can in turn generate benefits such as making it easier for consumers to find the products they are looking for and ascertain the quality of those products. Accordingly, trademark law can create incentives for firms to invest in producing goods and services of consistently high quality. In turn, this means that—for better or worse—trademark law often allows producers to charge supracompetitive prices without using patent law. Consider, for example, the higher price charged for Tylenol than for store-brand acetaminophen, even though the products are chemically identical. The brand name is worth something to consumers, who associate it with high quality (and diminished headaches).

The sources of U.S. trademark law include the federal Lanham Act at 15 U.S.C. § 1051 et seq. and state law. As with copyright law, registration is not required for federal trademark protection; trademark rights are established through use of the mark in commerce. But registering a trademark with the USPTO provides benefits such as putting other firms on notice of the mark, establishing nationwide priority, and making it harder to argue that the mark is invalid.

Trademark law can protect “any word, name, symbol, or device” used “to identify and distinguish” the source of goods or services. 15 U.S.C. § 1127. Courts have interpreted this coverage expansively to include aspects of product design, including color, shape, sound, and even fragrance (collectively referred to as “trade dress”). If consumers would immediately view a mark as signifying



the source of the good or service, such as for coined terms like “Zappos” shoes, it is deemed “inherently distinctive” and is immediately protectable. Other marks are only protectable once they have “acquired distinctiveness,” meaning that consumers have learned to view it as identifying source. For example, “American” airlines could initially be viewed as signifying a generic U.S. air carrier, but over time it has acquired distinctiveness and come to be viewed as signifying the brand American Airlines. Product design features can also gain acquired distinctiveness, as in the three chimes that signify NBC, the red shoe soles that signify Louboutin (at least among those who purchase \$800 shoes), the purple color of Nexium acid reflux pills, and product shapes of everything from Lego minifigs to Pepperidge Farm Milano cookies.

Like copyright law, trademark law attempts to channel IP protection for useful product features to utility patent law by denying protection to “functional” features, which can include features that make the product more useful or that make it more aesthetically pleasing. *TrafFix Devices v. Marketing Displays*, 532 U.S. 23, 32 (2001). As with copyright’s useful article doctrine, the dividing line between functional and non-functional trade dress has proven difficult to police. For instance, this question was at issue in litigation over Louboutin’s lacquered red sole: did consumers like the red sole because it signified that the shoe was made by Louboutin, or did they like it because it’s visually striking and attractive? (The court sided with Louboutin and held that it was the former.)

A valid trademark provides protection against anything that would create a “likelihood of confusion” on the part of consumers. Likelihood of confusion is evaluated using multi-factor tests that vary by federal appellate circuit. (Speaking of likelihood of confusion!) Courts have held that confusion is more likely when the mark is strong, when the defendant’s use is more similar to the mark, and when the defendant’s goods or services are more proximate to the plaintiff’s. If the mark is “famous” (think Coca-Cola or Louis Vuitton), trademark also protects against “dilution,” which is the use of similar marks on dissimilar goods or services. You can’t put the Coca-Cola logo on a lawnmower, even though Coca-Cola doesn’t make lawnmowers, because a consumer might think that Coca-Cola had endorsed the lawnmower and lower her opinion of the brand as a result.

Similar to copyright fair use, trademark law has its own outlets for a variety of speech interests, including interests in accurately describing one’s own product and interests in commenting on trademarked product. *Descriptive fair use* allows a third party to use a descriptive mark in good faith to describe *their own* goods or services—so even though American Airlines is a strong mark, a different airline may describe themselves as “an American air carrier.” There are also a variety of doctrines that allow a third party to use a mark to refer to the *mark owner’s* product for purposes like comparative advertising, reporting, parody, or artistic expression. For example, Pepsi can use the “Coca-Cola” mark when explaining that its soft drink tastes better, a newspaper can use the name of a music band when polling readers about their favorite band member, and Aqua can use Mattel’s “Barbie” trademark in the title and

lyrics of its “Barbie Girl” song. However, the Supreme Court recently limited the circumstances in which expressive uses of a trademark will be permitted in a case involving a “Bad Spaniels” squeaky dog toy designed to parody a bottle of Jack Daniel’s whiskey. *Jack Daniel’s v. VIP Products*, 143 S. Ct. 1578 (2023).

6. Recap, Remedies, and International Law

The following table summarizes the five areas of IP described above based on the key issues of what it means to have a valid IP right, when that right is infringed, and whether there are any important limitations or defenses.

	Validity	Infringement	Limitations
<i>Trade Secret</i>	secrecy (not generally known or ascertainable), value due to secrecy, reasonable efforts to maintain secrecy	protection against misappropriation: acquisition by improper means or in violation of confidential relationship	independent discovery, reverse engineering, disclosure
<i>Utility Patent</i>	novelty (relative to expansive prior art), nonobviousness, enablement for person having ordinary skill in the art (PHOSITA); <i>no laws of nature or abstract ideas</i>	rights to exclude others from making, using, selling, offering to sell, or importing the invention as defined by every element of the claim read by PHOSITA	limited experimental use rights, limited term (20 years from filing)
<i>Design Patent</i>	novelty, nonobviousness, disclosure through drawings, ornamentality (non-functionality)	same rights as utility patents; infringement based on whether ordinary observer familiar with prior art would view the designs as substantially the same (narrow scope)	limited term (15 years from grant), potentially limited scope
<i>Copyright</i>	independent creation, modicum of creativity, fixation; <i>no ideas, facts, or useful articles</i>	rights of reproduction, derivatives, distribution, performance, display; infringement based on actual copying and substantial similarity	fair use, independent creation, not-very-limited term (can be >100 years!)
<i>Trademark</i>	source-identifying, distinctiveness, priority of use in commerce; <i>no generic words or functional features</i>	protection against likelihood of consumer confusion or dilution (for famous marks) due to defendant’s use in commerce	descriptive or nominative fair use

A final important issue across all areas of IP is what the *remedy* is when an IP right is infringed. In every area, the IP owner may seek an *injunction*, a court order forcing the infringer to stop the infringing behavior. Whether an injunction will be granted is based on factors described by the Supreme Court in *eBay v. MercExchange*, 547 U.S. 388 (2006), including whether money damages are sufficient to compensate for the injury. For example, a “patent assertion entity” that makes no products itself is likely to receive damages rather than an injunction, whereas a small startup that will be put out of business by the infringing behavior is more likely to be awarded an injunction. IP owners can also receive money damages for all types of IP infringement, including statutory damages—damages that do not require any harm to be proven—for copyright infringement and for trademark counterfeits. Criminal liability is also a possibility in trade secret, copyright, and trademark law.

Although this section has focused on U.S. law, note that international treaties have harmonized IP law across countries to a greater degree than for many other areas of substantive law. In particular, most countries belong to the World Trade Organization (WTO) and are thus bound by the 1994 Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS), which requires all but the least-developed nations to protect IP rights of all other TRIPS members at or above a basic level and to treat foreigners no less favorably than locals. Thus, a German or Chinese inventor can obtain the same U.S. IP rights as an inventor based in the United States, and a U.S. firm can similarly obtain IP rights in other countries. When might it make sense for a domestic firm to seek foreign IP protection?

C. Case Studies

The following case studies place you in the position of intellectual property counsel to a business. Your goal in each situation is to determine the best strategy that the firm you represent should pursue to maximize its own market success. You should endeavor to describe a full IP strategy for your firm, including:

1. Of the five categories above, what IP rights should the firm pursue and over what aspects of the firm’s business? (Remember that one product is often protected by more than one type of IP.)
2. What are the likely major hurdles to acquiring and protecting those rights, and how should the firm go about trying to overcome those hurdles?
3. What risks exist regarding others’ IP rights, and how should the firm best address those risks?

Consider these questions not just from a legal perspective but also from a *business* perspective. In addition, you should remove yourself from the role of IP counsel, place yourself in the role of policymaker, and think more systemically about the issues

presented: how do you think IP law *should* treat each of these cases? (These cases are drawn from real-life examples that may be easily recognizable, but for purposes of this exercise, please ignore any outside knowledge you have of the companies involved.)

Case 1: Magnetic Building Blocks

You are IP counsel to a company that manufactures plastic toy building blocks. These blocks have small magnets inserted in them, so they “stick” together without having to be physically connected. To your knowledge, your firm is the first to have developed toy blocks that connect only via magnets, though the idea of connecting objects using magnets is hardly a new one. Nonetheless, you are aware of a number of other firms that are looking to begin manufacturing plastic toys of this type. Your firm is worried about becoming merely one seller in a very crowded marketplace and is considering ways of distinguishing itself. In addition to making blocks in primary colors and allowing customers to construct their own creations, the firm has considered producing sets of blocks that will allow the user to create iconic fictional structures—a Green Gables set, a Royal Palace of Wakanda set, a Pemberley set, etc.

Case 2: Online Marketplace

You are IP counsel to a tech startup that is looking to create a new type of online marketplace that will link individual buyers and sellers. The marketplace will automatically monitor the conduct of buyers and sellers via its online platform. It will be able to track how quickly a buyer submits payment and how reliable that method of payment is; how quickly the seller mails the good being sold and how soon it arrives; whether the condition of the good matches the advertisement for the good (based on photos taken at both ends of the transaction); how responsive the buyer and seller are to one another’s emails; and so forth. This will provide a measure of each individual’s reliability without the need for anyone to write a review or give a rating.

At bottom, the startup’s invention consists of two components: (a) the idea of a peer-to-peer marketplace with reliability established through automation and photos; and (b) the lines of computer code that will actually make the marketplace work via a smartphone app. In addition, the startup is aware that there are a variety of pre-existing patents on various types of online marketplaces, most notably a patent that includes a claim for “an online peer-to-peer marketplace in which the trust is built through repeated interactions between users.”

Case 3: Record Label with a Likely Hit

You are the lead IP strategist for a major record label. One of the label’s biggest stars has just produced an absolute banger of a song that is likely to be a huge hit.

However, the star said she was inspired to write the song by another famous pop song that was released approximately thirty years earlier by a pop star who is now deceased. The melodies aren't the same, but the songs have some commonalities that a musicologist might recognize. The estate of the earlier pop star has indicated a willingness to license their IP when asked. In addition, the earlier song actually originated with another (non-famous) musician *forty* years ago. This musician sued the now-dead pop star when his song was first released (thirty years ago), and they settled for an undisclosed sum rumored to be in the millions of dollars. What is more, your star's new song also has several lyrics that are very similar to another famous song from approximately thirty years ago. Your star says she is quoting these lines to illustrate the absurdity of songwriting in that decade.

At the same time, you expect this song to be a big enough hit that you'll be able to engage in a lot of cross-promotional marketing around it. Your label might be able to sell merchandise with the name of the song or lyrics from the song printed on it. The star might also want to develop related lines of business in which she sells products that are loosely connected with this song.

Case 4: Vaccine Against a Global Disease

You are IP counsel to a major pharmaceutical manufacturer. Your firm is currently in the process of developing a vaccine that will help prevent a significant global disease. Accordingly, you expect that this vaccine, once developed, will have a huge market and earn your firm significant profits. However, the vaccine uses a new technology that was primarily developed by one of your firm's competitors. Instead of a synthetic molecule developed in a laboratory, the vaccine is based on an RNA strand of the disease that the vaccine is meant to combat. When patients are injected with a modified RNA strand of the disease, the body naturally has a strong immune response, even though the actual disease pathogen is not present. The competitor firm was the first to develop the idea of using RNA in this fashion, and it holds several early patents on the use of RNA to make vaccines. Your scientists say there are some differences between the technology developed by your competitor and how your firm is making the vaccine, though it is not clear how legally significant those differences are. It is also currently unknown whether the competitor firm has filed for patents on vaccines for this particular disease.

Another wrinkle is that the competitor firm has announced that it will not enforce its patents related to vaccines "against any firm that is developing a vaccine against [this disease] so long as the disease remains a significant global issue." This announcement was made in a press release and has not been formalized in any other manner.